

Smart Greenhouse Controller Shopping List

Leaf & Logic

TEAM MEMBERS: David, Raj and Jordie

ITEM	LINK	COST	NOTES
Greenhouse	https://www.amazon.ca/Greenhouse-23x12x16-5-	\$35.99	
LED Lights	https://www.amazon.ca/Garpsen-Lights-Spectrum-	\$23.99	Currently have a \$4/off Coupon on
USB Speaker	https://www.amazon.ca/Computer-Speakers-NJSJ-	\$18.99	
Hose	https://www.homedepot.ca/product/rain-bird-1-2-	\$17.48	
Irrigation Fitting Kit	https://p-nt-www-amazon-ca-	\$17.27	Currently have a \$1/off Coupon on
Wood base	https://www.homedepot.ca/product/alexandria-	\$18.48	
Water Pump	https://www.amazon.ca/Jebao-PP-333LV-	\$20.34	
Screws	https://www.homedepot.ca/product/paulin--10-x-3-	\$13.10	\$1.17ea x 10
Glue	https://www.amazon.ca/Gorilla-Non-Foaming-	\$14.57	
Conduit Straps	https://www.homedepot.ca/product/iberville-two-	\$3.73	
Micro SD	https://p-nt-www-amazon-ca-	\$14.31	
Fan	https://www.amazon.ca/NF-A14-iPPC-3000-PWM-	\$36.90	
Bucket	https://www.homedepot.ca/product/the-home-	\$3.47	
Bucket Lid	https://www.homedepot.ca/product/hdg-19l-5-gallon	\$3.27	
Raspberry Pi Tall Header	https://bc-robotics.com/shop/raspberry-pi-tall-header	\$3.95	
3x4 Right Angle Male Header – 4 Pack	https://bc-robotics.com/shop/3x4-right-angle-male-header	\$4.95	
Analog to Digital Converter	https://bc-robotics.com/shop/16-channel-analog-input	\$14.95	*ONLY 2 LEFT IN STOCK
	Subtotal	\$265.74	
	Tax	\$31.89	
	Total	\$297.63	

ITEMS WE ALREADY HAVE
Raspberry Pi 4
Raspberry Pi Protection Case
Soil Moisture Sensor
Temperature Sensor
Humidity Sensor
Relay Module
Breadboard
Light Sensor
Voltage Converter
Small Wires
Jumper Wires (male to female)

Existing Knowledge and Expertise:

- Web Development Skills: Knowledge of programming, particularly in languages and frameworks relevant to Raspberry Pi (Python, JavaScript, etc.).
- Plant Care Expertise: Understanding of plant needs, growth conditions, and care.

Software and Development Tools:

- Raspberry Pi OS: The operating system for the Raspberry Pi, available for free download.
- Integrated Development Environment (IDE): Software for writing and testing code, such as Visual Studio Code, which is free to use.
- Version Control System: Tools like Git, with platforms such as GitHub or GitLab for code storage and collaboration, available in free versions.

Online Resources and Documentation:

- Raspberry Pi and Sensor Documentation: Official documentation and community forums.
- Open-Source Libraries: For interfacing with sensors, controlling actuators, or designing UIs.
- Energy Efficiency Guidelines: Online articles, guides, and research papers on optimizing power usage.

Existing Hardware and Components:

- Personal or Shared Computers: For software development, testing, and interfacing with the Raspberry Pi.
- Networking Equipment: Routers or modems for internet connectivity, essential for remote monitoring features.
- Power Supplies: Chargers and cables that come with the Raspberry Pi and other components.
- Miscellaneous Electronic Components: Wires, breadboards, resistors, and capacitors that might already be available in your toolkit.

Collaboration Platforms:

- Project Management Tools: Platforms like Trello or Asana, which offer free tiers for managing tasks and deadlines.
- Communication Tools: Slack, Discord, or Microsoft Teams for team discussions, many of which have free access.

DIY Tools and Supplies:

- Basic Hand Tools: Screwdrivers, pliers, and cutters for assembling the greenhouse structure.
- Measuring Tools: Rulers, measuring tapes, and possibly digital calipers for precise measurements.

Recycled or Repurposed Materials:

- Containers or Pots: Previously used pots or containers that can be repurposed for the greenhouse.
- Structural Materials: Any reusable wood, plastic, or metal that can be utilized in constructing the greenhouse base or frame.